

Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and computer vision: Gaussian splatting pipelines, benchmarking systems that measure every change, and continuous research that turns the best new papers into production code. Financial Engineering student at RTU.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

Solved blurry windows and mirrors in sparse-panorama 3DGS. instagram.com/p/DZsNRB8o772
This was a long-standing quality problem. I found the root cause (wrong depth behind windows and walls), handled depth differently inside and outside the room, and added a training-time correction. 700+ controlled experiments back it. It runs in production and was publicly showcased by Giraffe360.

Shipped the looped drone video, from idea to production. instagram.com/giraffe360/reel/Dd9ODBVIHhE
I proposed the feature, agreed its look with the designers, developed the camera-path algorithm and shipped it to production. A customer-requested feature, launched by Giraffe360 in Oct 2026.

Led the team that built outdoor splats and orbital videos.
Led the team that built exterior (outdoor) splats and the orbital outdoor videos, now in production.

Research to production: influence accumulation, plus Analytic-Splatting in gsplat.
Influence accumulation measures each Gaussian's contribution to the image and relocates the least useful ones during training; it is in production. Analytic-Splatting (exact per-pixel Gaussian integration) is implemented in gsplat at CUDA level. It reduces aliasing and is used in a downsampling algorithm.

Built tools to measure 3DGS quality against known ground truth.
Validation tools and ground-truth datasets, to measure quality against known truth; an experiment runner for agent-driven research loops on local GPUs; ClearML GPU runs.

Mentored a junior engineer on a new depth algorithm.
The junior engineer built a new depth algorithm with me (mostly his work): minima matching plus OMNI-DC to densify LiDAR depth. Now the best method; the team uses it.

Led the pipeline restructuring and created a 3DGS monorepo.
Led the pipeline restructuring and created a monorepo for all 3DGS extensions and client apps: ~3000-line files split, duplication removed, defects fixed. Enabled the half-floor camera fix.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.
Embeddable Rust/wgpu splat viewer (host control, PLY comparison); multi-view comparison tool.

Prototyped mirror planes and a fly-in dollhouse video intro.
Working prototypes: mirror planes with correct reflections; a fly-in "dollhouse" video intro.

Worked on mesh generation and point-count reduction for viewers.
Mesh generation from splats, combined with Gaussian splats; point-count reduction for viewers.

Made diffusion-based artifact fixing runnable on shared GPUs.
Diffusion-based artifact fixing (Difix, NVIDIA Fixer) made runnable on shared GPUs.

Three hackathon ideas recognized; the looped drone shipped.
3 ideas recognized at an internal hackathon (top 20, 2 mentions); the looped drone became a shipped feature.

Built an MCP server for the team task board.
An MCP server so AI agents can read and write the team task board (Node.js).

Earlier: a small LLM service that wrote tour narration.
Jul–Oct 2025: a small LLM service that wrote tour narration (OpenAI API, AWS, S3).
Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering 2024 – 2028 (expected)

Maths, statistics and programming applied to finance.

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

AI Talent Program (Ponder) Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

Selective program in C++, Python, maths and AI.

Selective program: C++, Python, maths and AI development with industry mentors. Led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

Rīgas Valsts klasiskā ģimnāzija

State mathematics olympiads and the Small Mathematics University.

Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

load-balancing-experiments — seven experiments on DNS, queueing and load balancing.

flow — offline workout tracker in Dart/Flutter, in daily use.

monte-carlo-finance — option pricing, volatility and risk models in Python.

Portfolio journey — this tap-driven WebGL site in three.js.

Offline Graphing Calculator — 4th place of 27 in a school competition.

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

Snake + maze generator — maze generation, terrain and path-finding in Pygame.

bezier-curve-js — interactive Bezier-curve drawing tool.

sort_visualization — 16 sorting algorithms animated in C++/SFML.

SKILLS

3D & vision	Gaussian splatting, 3D reconstruction, computer vision, depth estimation, anti-aliasing, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA kernels
Research	Paper-to-production, benchmarking and evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, ClearML, AWS S3, Pygame, Selenium, pandas
Other	OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL
Spoken	Latvian, English, Russian